Quiz 1 - Week 1 Slides

Wednesday, March 4, 2026 6:07 PM

Question 1:
What is the primary role of an operating system?
- ◻ Execute application programs directly
- ◻ Manage hardware resources and provide abstractions for application programmers
- ◻ Compile user code into machine language
- ◻ Replace the need for a CPU

Question 2:
What is the primary difference between a process and a thread?
- ◻ Threads have their own address space, while processes share one
- ◻ Processes have their own address space, while threads share one
- ◻ Threads execute only in kernel mode
- ◻ Processes do not require memory allocation, but threads do

Question 3:
What is an example of time multiplexing in an operating system?
- ◻ A process using a shared memory segment for communication
- ◻ The CPU switching between different processes at rapid intervals
- ◻ Allocating separate memory blocks to each running application
- ◻ Assigning different I/O devices to different users

Question 4:
Which of the following statements about kernel mode and user mode is true?
- ◻ Kernel mode restricts access to CPU resources, while user mode allows full access
- ◻ User mode is where the OS executes all of its critical functions
- ◻ In kernel mode, the OS has full access to hardware and system resources
- ◻ Switching from kernel mode to user mode does not require a context switch

Question 5:
Which type of memory is used by the CPU for immediate processing and is the fastest?
- ◻ Hard Disk
- ◻ RAM
- ◻ Cache
- ◻ SSD

Question 6:
What are the key components of hardware that make up a modern computer system? Answer with a list of components as well a short description of the component.
Answer:

Question 7:
What are the core concepts that make up an operating system? Answer with a list of concepts as well as a short description
Answer: